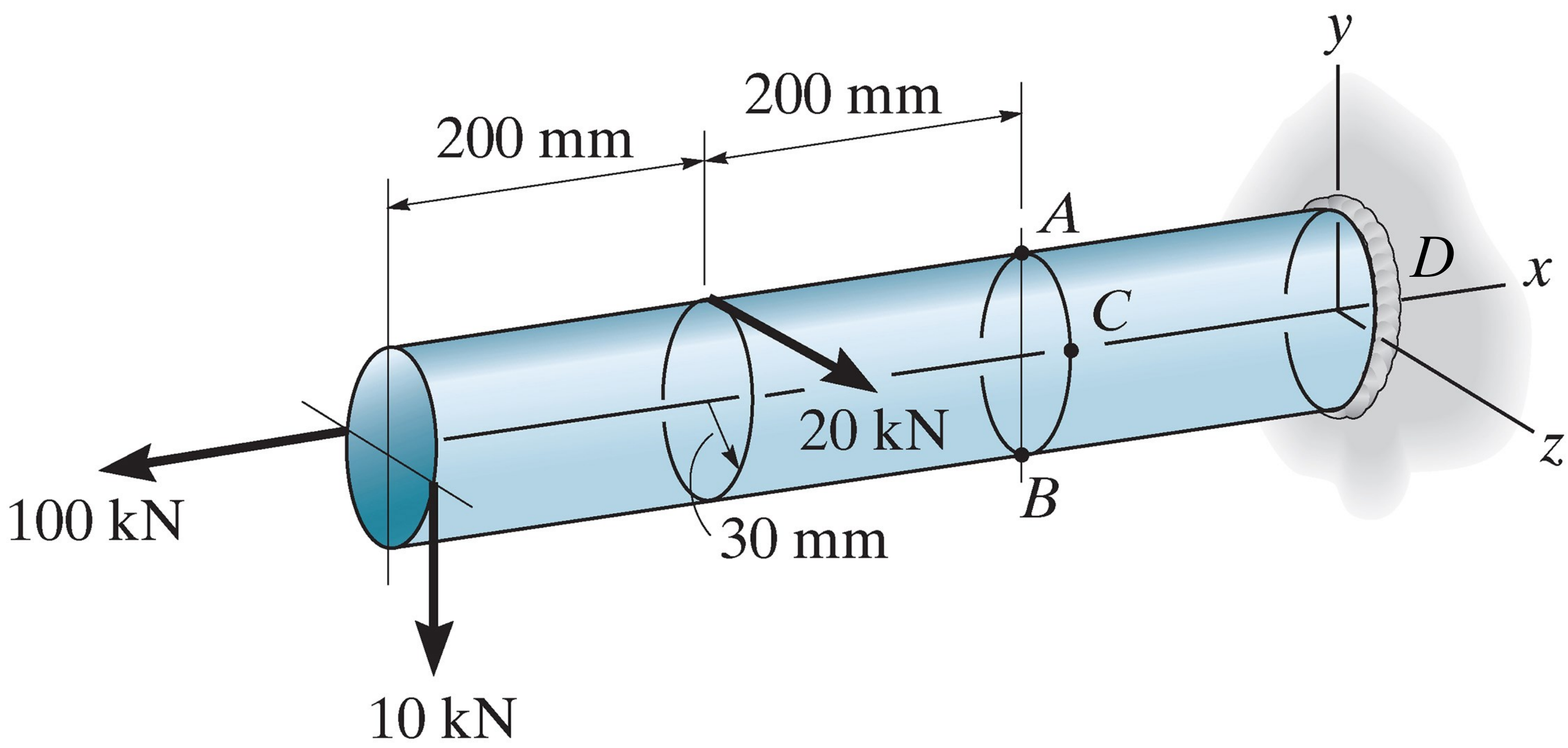
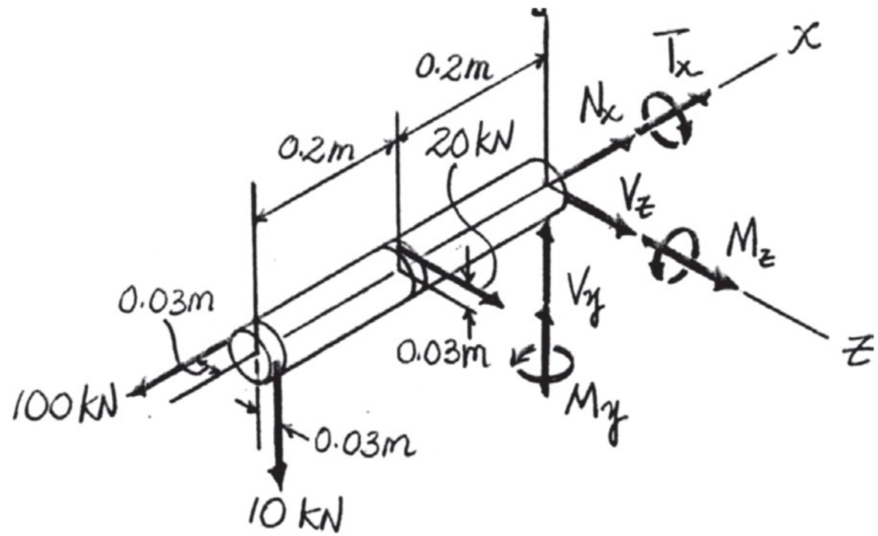
1. The 30-mm-radius rod (<u>note:</u> the cross-section is circular) is subjected to the loads shown. Determine the state of stress at point *A* (on positive *y*-axis). Also, show the results on a differential volume (representative volume of stress) located at the point *A*. The end *D* of the rod is fixed. (**20'**)



Solutions to Extra Exercise #1

Consider the equilibrium of the left segment of the rod being sectioned:



$\sum F_x = 0 \;\rightarrow\; N_x = 100\text{ kN}$

$\sum F_y = 0 \;\rightarrow\; V_y = 10\text{ kN}$

$\sum F_z = 0 \;\rightarrow\; V_z = -20\text{ kN}$

$\sum M_x = 0 \;\rightarrow$

$T_x + 20 \times 0.03 + 10 \times 0.03 = 0$

$T_x = -0.9\text{ kN}\cdot\text{m}$

$\sum M_y = 0 \;\rightarrow$

$M_y + 20 \times 0.2 + 100 \times 0.03 = 0$

$M_y = -7\text{ kN}\cdot\text{m}$

$\sum M_z = 0 \;\rightarrow$

$M_z + 10 \times 0.4 = 0$

$M_z = -4\text{ kN}\cdot\text{m}$

Solutions to Extra Exercise #1, cont'd

Section properties:

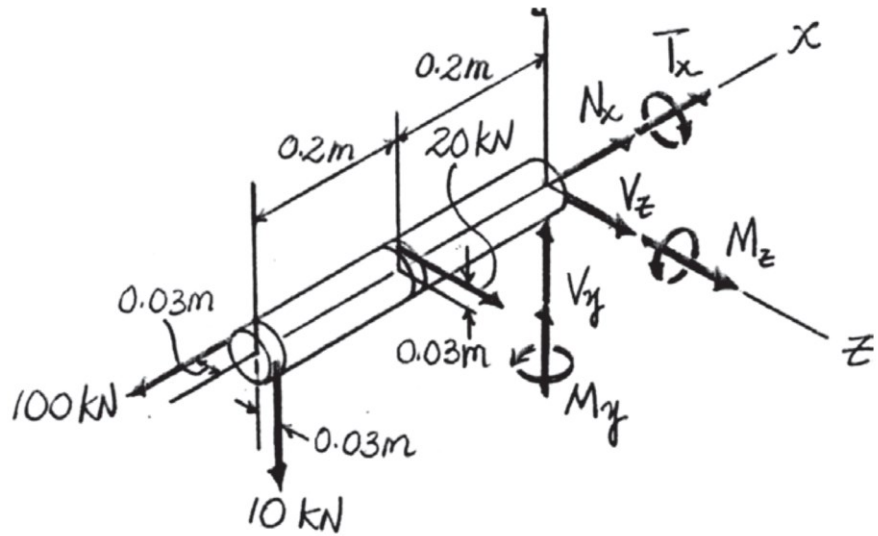
$$A = \pi c^2 = 2.83 \times 10^{-3}\ \text{m}^2$$

$$I_y = I_z = \frac{\pi}{4} c^4 = 6.36 \times 10^{-7}\ \text{m}^4$$

$$J = \frac{\pi}{2} c^4 = 1.27 \times 10^{-6}\ \text{m}^4$$

$$(Q_A)_z = \overline{z}' A' = 1.8 \times 10^{-5}\ \text{m}^3$$

$$(Q_A)_y = 0$$



For point A:

Normal Stress: For the combined loadings, the normal stress at point A can be determined from

$$\sigma_x = \sigma_A = \frac{N_x}{A} - \frac{M_z y_A}{I_z} + \frac{M_y z_A}{I_y} = 224\ \text{MPa} \quad \text{(Tension)} \qquad \textbf{(Ans.)}$$

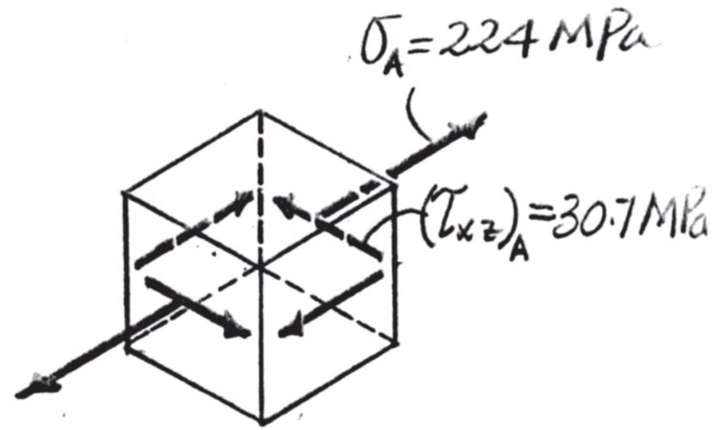
Solutions to Extra Exercise #1, cont'd

Shear Stress: The transverse shear stress in z and y directions and the torsional shear stress can be obtained using the shear formula $\tau_V = \frac{VQ}{It}$ and the torsion formula $\tau_T = \frac{T\rho}{J}$, respectively.

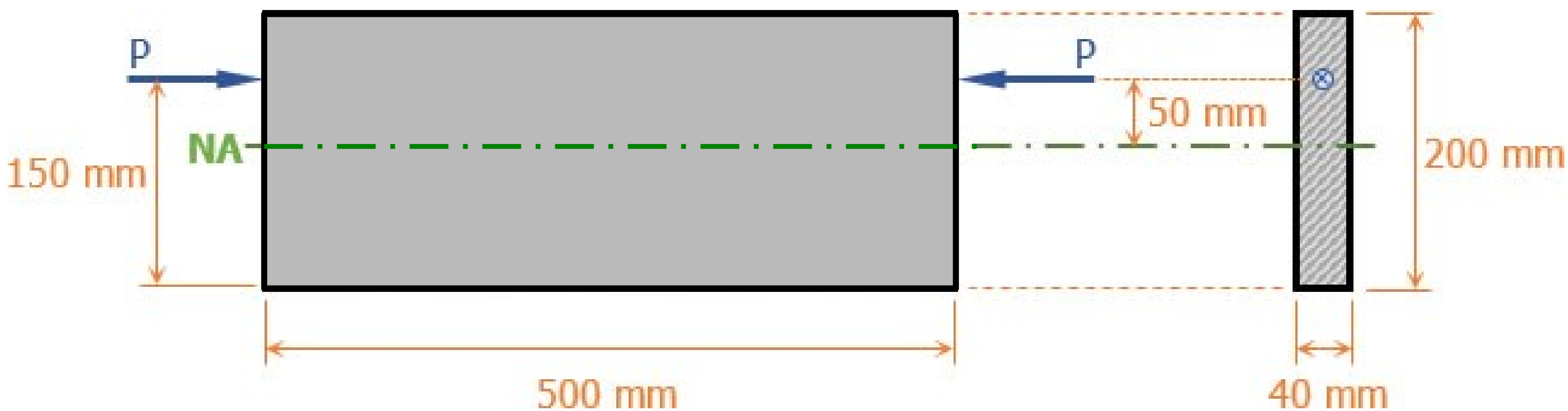
$(\tau_{xz})_A = (\tau_V)_z + \tau_T = -30.7$ MPa **(Ans.)**

$(\tau_{xy})_A = (\tau_V)_y = 0$ **(Ans.)**

Using these results, the state of stress at point *A* can be represented by the differential volume element shown on the right.



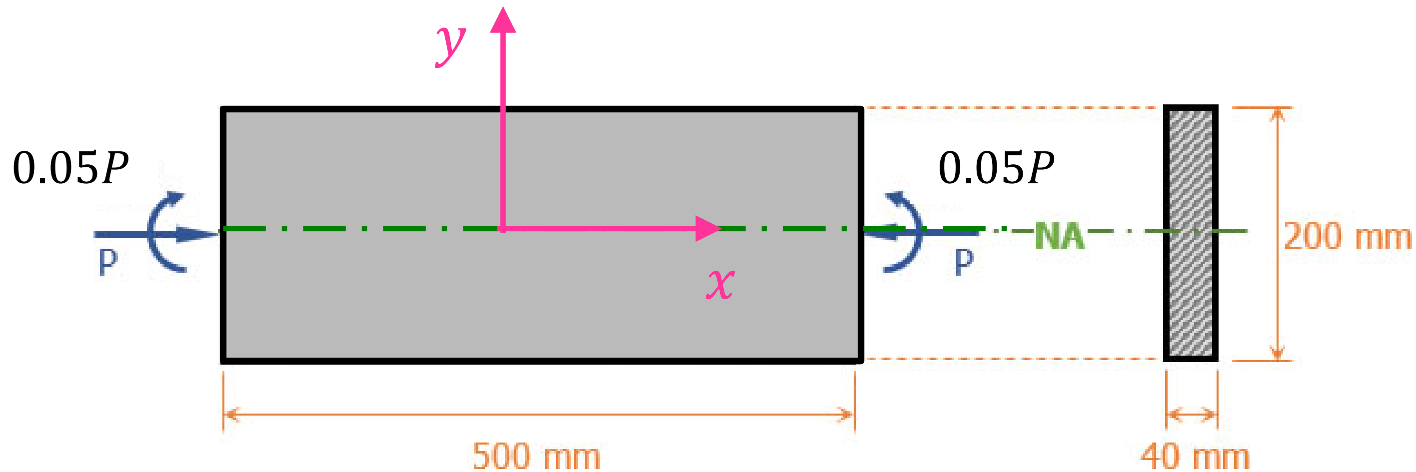
2. A cast iron link has dimension of 40 × 200 × 500 mm as shown in the figure. The allowable stresses are 40 MPa in tension and 80 MPa in compression. Compute the largest compressive load *P* that can be applied to the ends of the link along a longitudinal axis that is located 150 mm above the bottom of the link. (**20'**)



Solutions to Extra Exercise #2

Equivalent Loading

$M = 0.05P\ \mathrm{N \cdot m}$

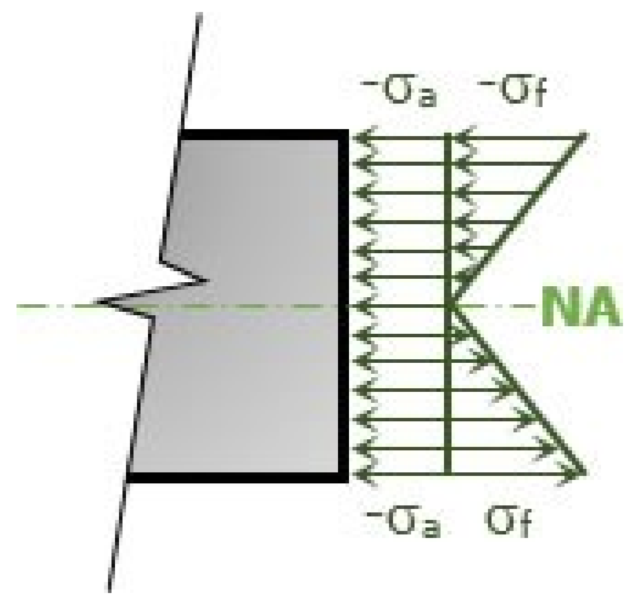


The normal stress developed is the combination of axial and bending stress.

For the top part of the link (compressive stress):

$$\sigma_{top} = -\sigma_a - \sigma_f$$

$$\sigma_{top} = -\frac{P}{A} - \frac{M \times h/2}{\frac{1}{12}bh^3}$$



Solutions to Extra Exercise #2, cont'd

To fulfill the condition for compression, we have

$$\sigma_{top} = -\frac{P}{A} - \frac{M \times h/2}{\frac{1}{12}bh^3}$$

$$\left|\sigma_{top} = -\frac{P}{A} - \frac{6M}{bh^2}\right| \leq \sigma_{allow}|compression = 80 \text{ MPa}$$

$$\frac{P}{0.04 \times 0.2} + \frac{6 \times 0.05P}{0.04 \times 0.2^2} = 80 \times 10^6$$

$\rightarrow$ $P = 256 \text{ kN}$

Check the bottom part of the link (tensile stress):

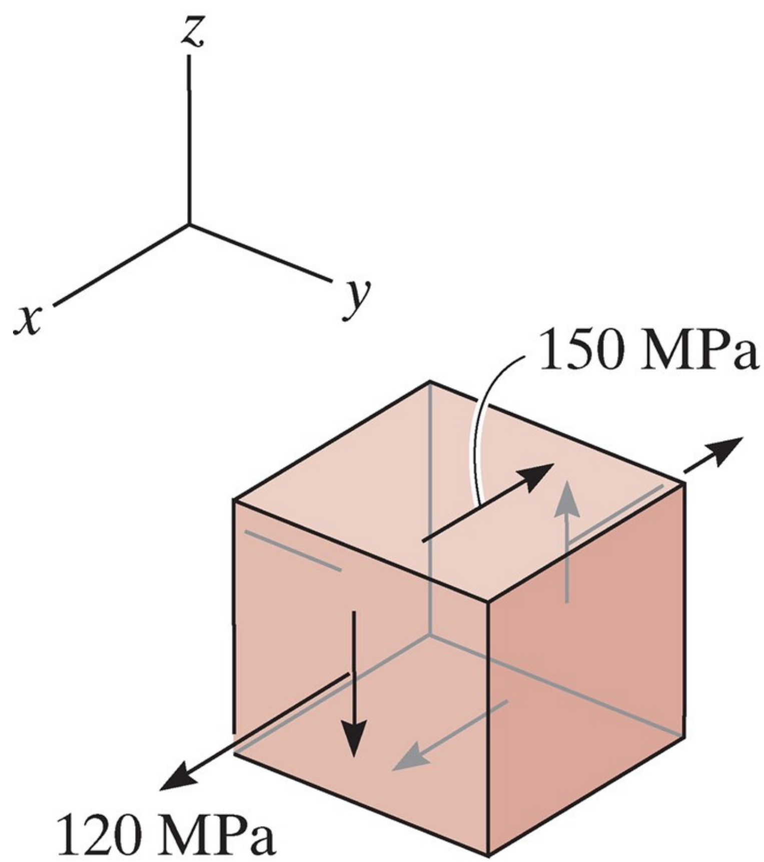
$$\sigma_{bottom} = -\sigma_a + \sigma_f$$

$$\sigma_{bottom} = -\frac{P}{A} + \frac{Mh/2}{I} = -\frac{256000}{0.04 \times 0.2} + \frac{6 \times 0.05 \times 256000}{0.04 \times 0.2^2} = 16 \text{ MPa}$$

$$\sigma_{bottom} < \sigma_{allow}|tension = 40 \text{ MPa} \quad \text{(O.K.)}$$

Finally, we get $P = 256 \text{ kN}$ **(Ans.)**

3. Due to an applied loading, an element at a point on a machine shaft is subjected to the state of plane stress shown below. Determine the principal stresses and the absolute maximum shear stress at the point. (**15'** points)



Solutions to Extra Exercise #3:

The two stresses occur in *xz*-plane. Thus, this is a plane-stress problem for *xz*-plane.

Given $\sigma_x = 120$ MPa, $\sigma_z = 0$, $\tau_{xz} = -150$ MPa, the center of the Mohr's circle *C* is located at the point *C*(60, 0) and the reference point *A*(120, -150) are labeled. Then, the radius of the circle is $R = \sqrt{(120-60)^2 + 150^2} = 161.55$, and the circle is constructed.

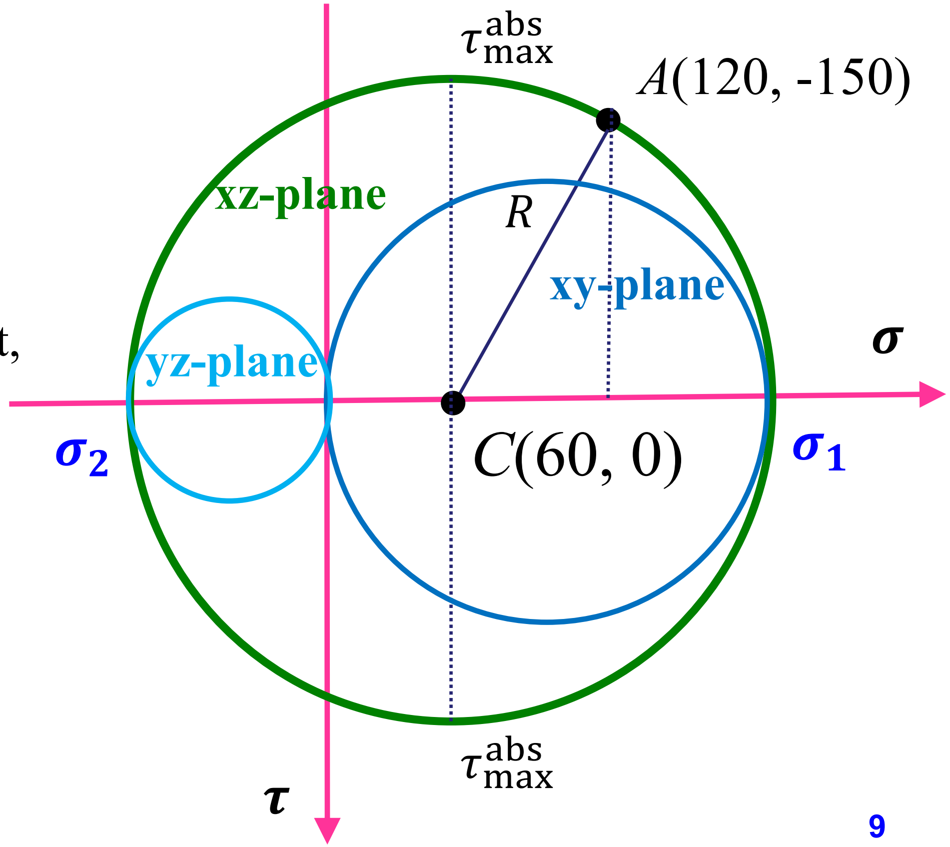
The principal stresses are:

$\sigma_1 = 60 + 161.55 = 221.55$ MPa **(Ans.)**

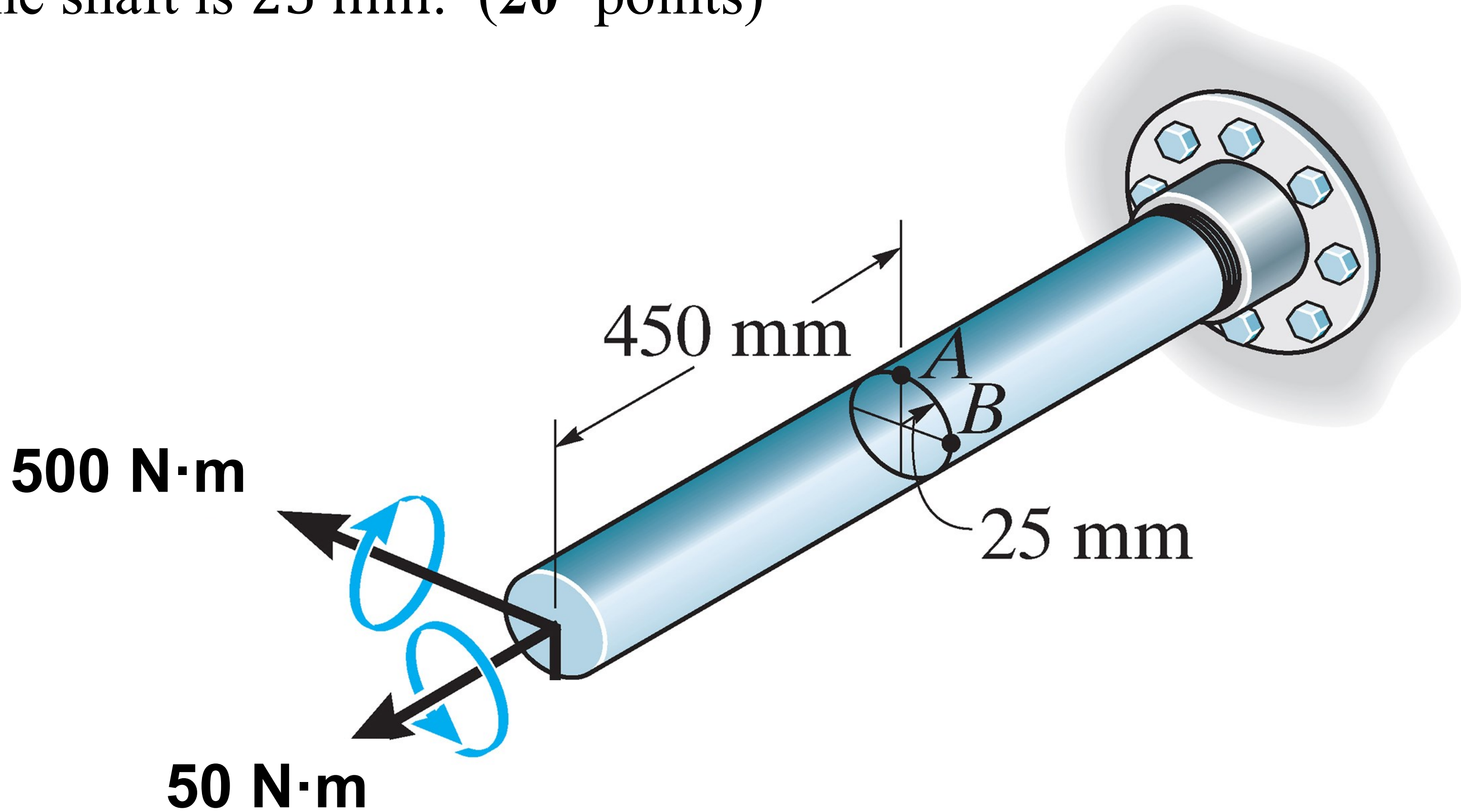
$\sigma_2 = 60 - 161.55 = -101.55$ MPa **(Ans.)**

By drawing three Mohr's circles, we know that, the absolute maximum shear stress is:

$$\tau_{\max}^{\text{abs}} = \frac{\sigma_1 - \sigma_2}{2} = 162 \text{ MPa}$$ **(Ans.)**



4. The solid shaft is subjected to a torque and a bending moment. Determine the principal stresses at points *A* and *B*, and the absolute maximum shear stresses at these points. The radius of the shaft is 25 mm. (**20'** points)



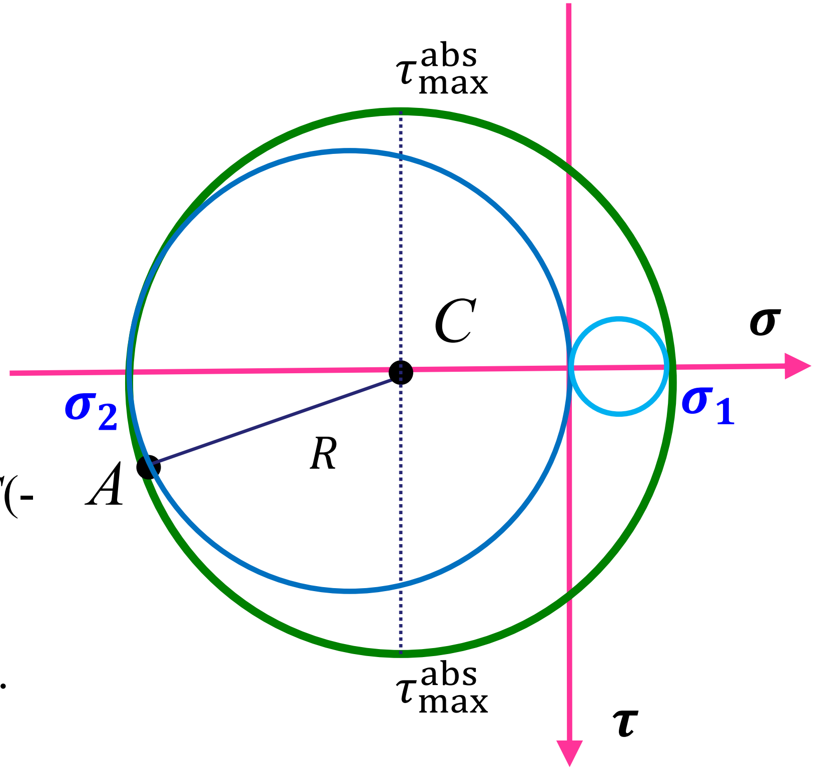
Solutions to Extra Exercise #4

For point A:

The bending moment induces compressive normal stress, while the torque induces shear stress.

$$\sigma_x = -\frac{M \cdot y}{I} = -\frac{500 \times 0.025}{\frac{1}{4}\pi \cdot 0.025^4} = -40.7 \text{ MPa}$$

$$\tau_{xy} = \frac{Tc}{J} = \frac{50 \times 0.025}{\frac{1}{2}\pi \cdot 0.025^4} = 2.0 \text{ MPa}$$



Given $\sigma_x = -40.7$ MPa, $\sigma_y = 0$, $\tau_{xy} = 2$ MPa, the center of the Mohr's circle C is located at the point C(-20.35, 0) and the reference point A(-40.7, 2) are determined. Then, the radius of the circle is $R = \sqrt{20.35^2 + 2^2} = 20.45$, and the circle is constructed.

Principal stresses:

$\sigma_1 = 0.1$ MPa, $\sigma_2 = -40.8$ MPa **(Ans.)**

The absolute maximum shear stress: $\tau_{\max}^{\text{abs}} = 20.45$ MPa **(Ans.)**

Solutions to Extra Exercise #4, cont'd

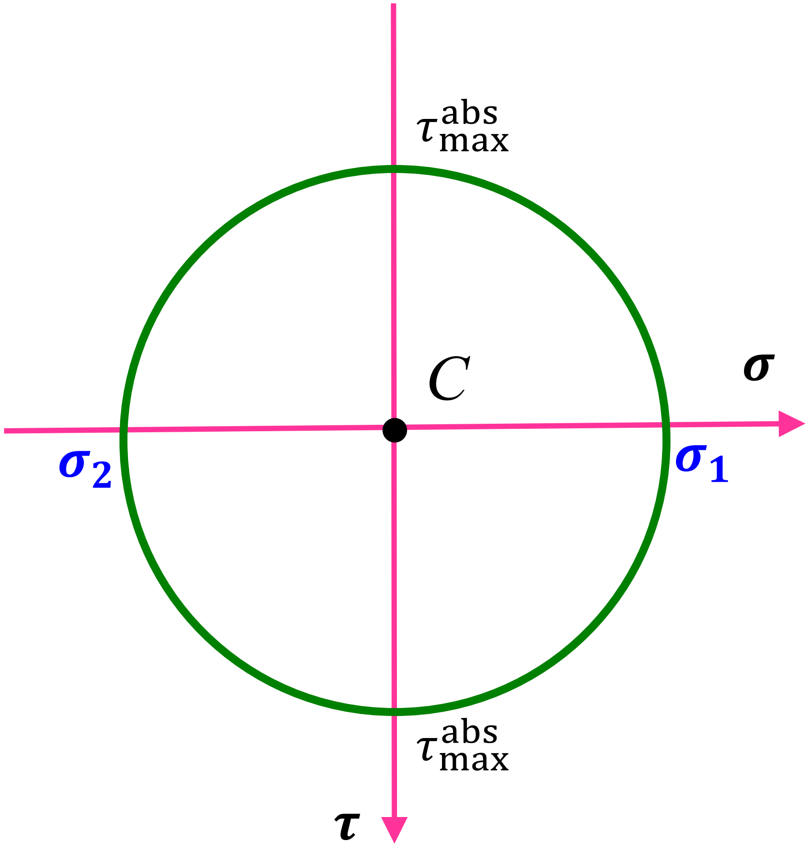
For point B:

The bending moment does not induce normal stress, while the torque induces the same shear stress.

$$\sigma_x = \sigma_y = \sigma_z = 0$$

$$\tau_{xy} = \frac{Tc}{J} = \frac{50 \times 0.025}{\frac{1}{2}\pi \cdot 0.025^4} = 2.0 \text{ MPa}$$

Therefore, this is a pure shear problem.
The corresponding Mohr's circle is shown.



The radius of the circle is
$R = 2$

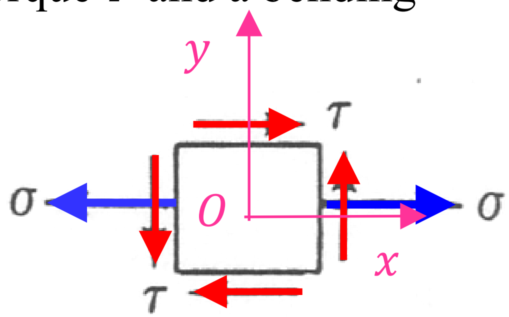
Principal stresses:
$\sigma_1 = 2$ MPa, $\sigma_2 = -2$ MPa **(Ans.)**

The absolute maximum shear stress: $\tau_{\max}^{\text{abs}} = 2$ MPa **(Ans.)**

5. A cylindrical shaft is subject to a torque T and a bending moment M. Express the maximum shear stress τ_{max} and the maximum normal stress σ_{max} in terms of T, M, and the radius of the shaft (r). Using this relation, determine the proper diameter of a solid shaft to carry simultaneously $T = 1220\ \text{N} \cdot \text{m}$ and $M = 810\ \text{N} \cdot \text{m}$ if $\tau \leq 70\ \text{MPa}$ and $\sigma \leq 110\ \text{MPa}$ anywhere in the shaft. (**25'** points)

Solutions to Extra Exercise #5

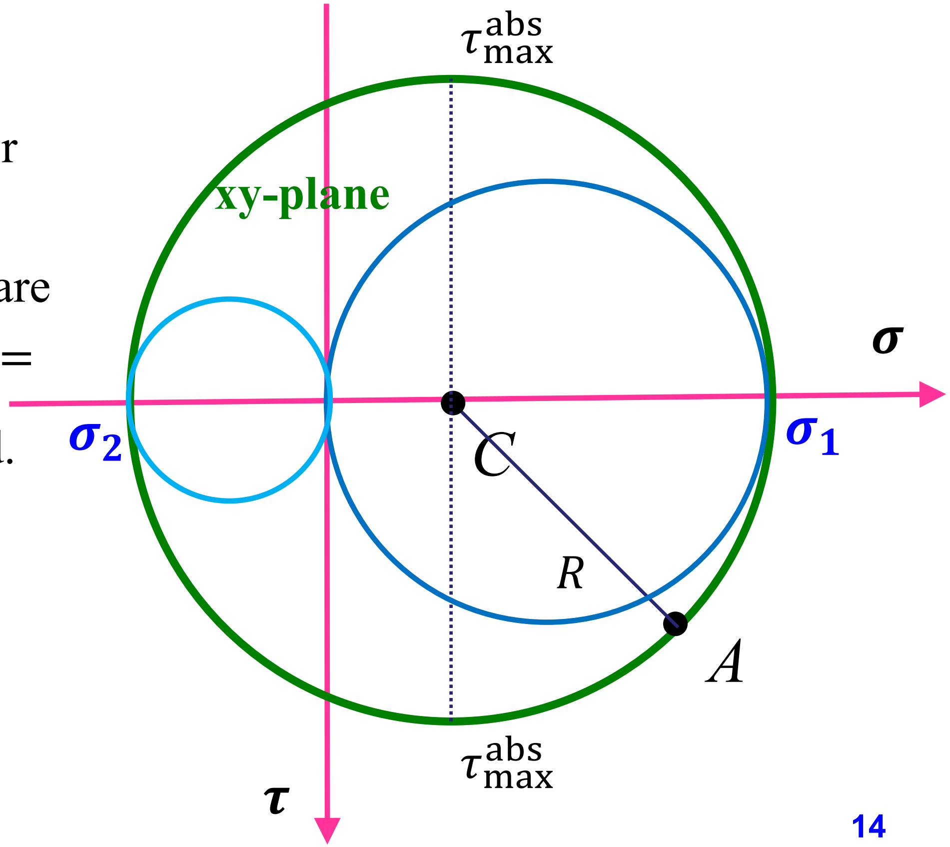
For any point on the cylindrical shaft that is subject to a torque T and a bending moment M, the stress state can be represented by:



$$\sigma = \frac{Mc}{I} = \frac{Mr}{\frac{\pi}{4}r^4} = \frac{4}{\pi}\frac{M}{r^3}$$

$$\tau = \frac{Tc}{J} = \frac{Tr}{\frac{\pi}{2}r^4} = \frac{2}{\pi}\frac{T}{r^3}$$

Given $\sigma_x = \frac{4}{\pi}\frac{M}{r^3}$, $\sigma_y = 0$, $\tau_{xy} = \frac{2}{\pi}\frac{T}{r^3}$, the center of the Mohr's circle C is located at the point $C(\frac{2}{\pi}\frac{M}{r^3}, 0)$ and the reference point $A(\frac{4}{\pi}\frac{M}{r^3}, \frac{2}{\pi}\frac{T}{r^3})$ are determined. Then, the radius of the circle is $R = \sqrt{(\frac{2}{\pi}\frac{M}{r^3})^2+(\frac{2}{\pi}\frac{T}{r^3})^2}$, and the circle is constructed.



Solutions to Extra Exercise #5, cont'd

The principal stresses are:

$$\sigma_1 = \frac{2}{\pi}\frac{M}{r^3} + R = \frac{2}{\pi}\frac{M}{r^3} + \sqrt{(\frac{2}{\pi}\frac{M}{r^3})^2 + (\frac{2}{\pi}\frac{T}{r^3})^2} \; > \; 0$$

$$\sigma_2 = \frac{2}{\pi}\frac{M}{r^3} - R = \frac{2}{\pi}\frac{M}{r^3} - \sqrt{\left(\frac{2}{\pi}\frac{M}{r^3}\right)^2 + \left(\frac{2}{\pi}\frac{T}{r^3}\right)^2} \; < \; 0$$

Thus,

$$\sigma_{\max} = \sigma_1 = \frac{2}{\pi}\frac{M}{r^3} + \sqrt{(\frac{2}{\pi}\frac{M}{r^3})^2 + (\frac{2}{\pi}\frac{T}{r^3})^2} \qquad \textbf{(1)}$$

$$\tau_{\max}^{\text{abs}} = \frac{\sigma_1 - \sigma_2}{2} = R = \sqrt{\left(\frac{2}{\pi}\frac{M}{r^3}\right)^2 + \left(\frac{2}{\pi}\frac{T}{r^3}\right)^2} \qquad \textbf{(2)}$$

With the condition $\tau \leq 70$ MPa and $\sigma \leq 110$ MPa anywhere in the shaft, $T = 1220 \text{ N} \cdot \text{m}$ and $M = 810 \text{ N} \cdot \text{m}$, we have:

Eq. (1) → $r \geq 23.6$ mm
Eq. (2) → $r \geq 23.7$ mm

Finally, the diameter of the shaft should be at least $2 \times 23.7 = \mathbf{47.4}$ **mm** **(Ans.)**